Majority Illusion in RAG -- Results & Discussion (draft)

Numbers are from the complete 75-entity standard run (1350 conditions, 3 models x 6 ratios). Replace point estimates with exact Wilson CIs / clustered p-values from run_all_analyses.py before submission.

3 Results

3.1 RQ1: Majority Dominance

The probability of selecting the majority claim rose steadily with the document ratio for all three models (Fig. 1). At the balanced 2:2 tie, majority selection was at or below chance (Claude 36%, DeepSeek 41%, Gemini 0%), and by the 4:1 ratio all three exceeded 99%. The conflict-free 4:0 control returned the unanimous value in 100% of conditions, confirming that behavior at the conflict ratios is driven by the disagreement rather than by task difficulty.

The transition was strongly model dependent. Claude and DeepSeek captured the majority almost immediately, exceeding 90% by the weakest majority condition (3:2), whereas Gemini resisted, holding near 57% through 3:2 and 2:1 before rising to 95% at 3:1. The full category breakdown (Fig. 4) shows the models use qualitatively different strategies at the tie: Claude and DeepSeek split between the two values (roughly 40% MAJ / 40% MIN), while Gemini never commits to a value (0% MAJ, 0% MIN), resolving every tied condition through compromise (52%) or flagging (48%).

3.2 Confidence and Conflict Awareness

Self-reported confidence in the chosen answer increased as the majority grew, from a mean of 79/100 at 2:2 to 91/100 at 4:1 (Fig. 3). Over the same range the independent estimate that the sources conflict remained high, declining only from 98 to 87. Awareness of disagreement was therefore near-total at every conflict ratio and was largely decoupled from behavior.

Figure 8 summarizes this dissociation. Reported conflict awareness stayed between 87% and 98% across all ratios, while the rate of protective action (explicit flagging) fell from 49% at 2:2 to 0% at 3:1 and 4:1, and majority-following rose in inverse lockstep from 19% to 100%. Among all majority-following conditions, the mean reported conflict probability was 92/100, and 78% reported a conflict probability of at least 50%.

3.3 RQ2: Domain Rigidity

Domain sensitivity produced no detectable change in caution (Fig. 7). Pooled across conflict ratios, banking entities (sensitive numeric financial parameters) and general-corporate entities yielded nearly identical majority-selection rates (77.6% vs 76.6%) and conflict-acknowledgement rates (FLAG 5.4% vs 6.8%). The intervals for the domain difference included zero for every outcome.

3.4 RQ3: Position Bias

Because minority position was assigned by deterministic shuffle rather than forced allocation, placements were unequal (first n=254, middle n=108, last n=179); this distribution is reported for transparency. Within the single-minority conditions (2:1, 3:1, 4:1), positional effects were small. Minority selection was slightly elevated when the minority document sat in the middle of the context (2.8%) relative to first (0.4%) or last (1.1%), and majority selection was correspondingly lowest in the middle (88.9% vs 95.7% first). No strong primacy or recency pattern emerged; the majority dominated regardless of placement.

3.5 RQ4: Effect of Confidence Elicitation

Requesting a structured probability distribution changed conflict-handling relative to the answer-only control (Fig. 6). Pooled across conflict ratios, the answer-only arm produced explicit abstention in 0.0% of conditions and expressed disagreement instead through compromise (COM 18.3%). The structured-elicitation arm reversed this: explicit flagging rose to 12.3% while compromise nearly vanished (1.6%), and majority selection increased modestly (74.2% to 79.7%). Confidence elicitation thus converted implicit compromise into explicit flagging rather than acting as a passive measurement.

3.6 Confidence Elicitation Reverses Direction Across Ratios (novel)

The effect of structured elicitation was not uniform; it reversed direction with the ratio (Fig. 9). At the 2:2 tie, elicitation reduced majority selection (31% to 19%, a change of -11 points) and sharply increased flagging (0% to 49%). At the slight-majority ratios the sign flipped: elicitation increased majority selection at both 3:2 (+12) and 2:1 (+19), with only small flagging changes. At the strong-majority ratios (3:1, 4:1), already saturated, it had negligible effect.

This crossover was accompanied by large heterogeneity across models. It was greatest in Gemini (flagging +95 at the tie; majority +38 to +43 at slight majorities), tie-specific in DeepSeek (majority -33, flagging +38 at 2:2, little elsewhere), and mild in Claude (flagging +11 at the tie; majority +14 at 2:1).

4 Discussion

4.1 Awareness without action

Our principal finding is that majority-following in RAG is a failure of resolution policy, not of perception. Across every conflict condition, models reported near-certain awareness that their sources disagreed, yet acted on it only at the balanced tie; once even a one-document majority was present, protective flagging collapsed while reported conflict probability stayed above 87% and confidence rose. This implies that interventions aimed at helping models notice conflict are misdirected -- they already notice -- and that confidence-gated safety filters cannot help, because confidence grows with mere repetition, precisely when a repeated-but-wrong claim is most dangerous. This extends the network-science majority illusion into retrieval, and goes beyond prior benchmarks (which show conflict is hard) by separating what the model knows from what it does.

4.2 Conflict resolution is a trained disposition, not a universal property

Given identical evidence, the three models resolved conflict in qualitatively different ways. Because the entities, documents, ratios, and ordering were held constant, these differences are attributable to the models. Conflict resolution is therefore best understood as a trained disposition that varies across independently developed systems, not a fixed property of language models. For practitioners this means model selection materially changes a RAG system's safety profile, and a pipeline validated on one model may behave very differently on another. We caution against inferring specific training mechanisms from behavior, as the developers' alignment procedures are only partly disclosed.

4.3 Confidence elicitation as an intervention

The arm contrast shows that how a model is queried changes what it does: eliciting structured confidence is an intervention, not a passive readout. Methodologically, this means measured abstention and flagging rates are prompt-dependent, so benchmarks that use a single prompt will misstate them. Practically, structured elicitation surfaces disagreement the model would otherwise resolve silently -- a cheap mitigation, but only under the conditions qualified in Section 4.7.

4.4 Positional effects and stability

Positional influence was minor: the lone minority document was marginally less influential when buried mid-context than at the edges, weakly consistent with prior lost-in-the-middle observations but overwhelmed by the frequency effect. The high self-consistency of responses (90-99%) indicates that the majority illusion is a systematic, reproducible disposition rather than stochastic error -- from a safety standpoint the less reassuring possibility, since it will recur predictably in deployment.

4.5 Insensitivity to domain stakes

Models showed no additional caution on sensitive financial parameters than on generic corporate facts. This null is consequential: it indicates that current models do not modulate conflict resolution by the stakes of the queried attribute. A system that treats a disputed interest rate exactly as it treats a disputed founding city will carry the same majority-illusion risk into high-stakes financial, legal, or medical retrieval, where the cost of confidently serving a repeated-but-wrong value is greatest.

4.6 Implications for RAG design

Taken together, these results point to concrete mitigations. Because models treat document count as evidence, pipelines should deduplicate at the level of distinct claims rather than documents, so a single repeated source cannot masquerade as corroboration. Because confidence rises with repetition, it should not be used as a safety gate. Because conflict resolution is model-specific, it should be a first-class criterion in model selection. And structured elicitation should be applied deliberately rather than as a blanket default (Section 4.7).

4.7 The Double-Edged Effect of Confidence Elicitation (novel)

The most striking property of confidence elicitation is that its direction depends on the balance of the evidence. When the evidence was tied, requesting a distribution surfaced the underlying uncertainty -- models abstained more and followed the majority less. But once even a slight majority existed, the same request had the opposite effect, crystallizing the model's small lean into confident majority-adherence. A plausible reading is that forcing an explicit allocation of probability amplifies whatever directional signal the context provides: at a tie there is none, so uncertainty is expressed as abstention, whereas a small majority supplies just enough signal for elicitation to consolidate rather than temper it. Because this pattern was dramatic in one model, tie-specific in another, and faint in a third, it is best understood as an interaction among elicitation method, ratio, and model. It cautions against treating confidence elicitation as a blanket safety measure: it can help at balanced evidence yet entrench majority-following exactly at the mildly imbalanced ratios where a repeated-but-wrong claim is most likely. To our knowledge this is the first controlled demonstration that structured confidence-and-conflict elicitation has ratio-dependent, directionally different effects under conflicting retrieved evidence.